

Degradation mechanisms and microstructure evolution of TGV-Cu vertical interconnects in glass wafer under high-temperature aging

Guoli Sun^{a,b}, Jinhong Liu^a, Jingyi Zhao^c, Jialin Zhang^a, Zhaoyu Li^d, Shuye Zhang^{a,b,*}

^a State Key Laboratory of Precision Welding & Joining of Materials and Structures, Harbin Institute of Technology, Harbin, 150001, China

^b Zhengzhou Research Institute of Mechanical Engineering, Zhengzhou, 450001, China

^c Institute of Microelectronics, Chinese Academy of Sciences, Beijing, 100029, China

^d China-Ukraine Institute of Welding, Guangdong Academy of Sciences, Guangdong Provincial Key Laboratory of Material Joining and Advanced Manufacturing, Guangzhou, 510650, China

ARTICLE INFO

Keywords:

TGV-Cu
Crack propagation
High-temperature aging
Microstructure evolution
Advanced packaging

ABSTRACT

This study investigates the failure mechanisms and microstructural evolution of through glass via-copper (TGV-Cu) interconnections under high-temperature storage conditions. Under aging conditions, the glass/Ti/Cu interface coarsens, leading to the formation of irregular voids and microcracks at the Cu/Ti interface that propagate under thermal stress and evolve into through-thickness cracks. The diffusion rate of Ti decreases as the system approaches thermodynamic equilibrium. Cracks predominantly propagate along the sidewalls of the TGV-Cu interconnections in a transgranular manner due to shear stresses at the glass/Ti/Cu interface. The grain size of TGV-Cu increases from an initial average of 3.12 μm to 9.97 μm , with a heterogeneous distribution. The initial preferred orientation is $\langle 001 \rangle$, but after aging for 50 h, shifts toward $\langle 101 \rangle$ and $\langle 111 \rangle$ directions, before reverting to $\langle 001 \rangle$ with further aging. Nanoindentation results indicate that the elastic modulus and hardness decrease initially, then increase during thermal aging, although overall performance remains lower than in the initial state. These findings provide insights into the integrity and lifetime prediction of TGV-Cu interconnections, aiding in the reliability design of glass-based electronic packaging systems.

1. Introduction

As the semiconductor industry enters the post-Moore era, driven by high-performance computing, large-scale models, and artificial intelligence, chip technology faces numerous challenges, including sharply rising costs, exponentially increasing computing demands, and bandwidth bottlenecks [1,2]. The traditional scaling benefits of Moore's Law are increasingly diminishing, and the performance enhancement path reliant on continuous shrinkage of chip feature sizes is no longer sufficient to meet the demands of cutting-edge applications [3]. In this context, advanced packaging technology has rapidly emerged as the core driver for continuing and even surpassing Moore's Law in the post-Moore era, becoming the key engine for industry development [4].

With high-density integration, three-dimensional stacking architectures, and excellent process compatibility, advanced packaging technologies provide strong support for the rapid evolution of cutting-edge application areas such as big data, intelligent robotics, and digital twins.

Among these, vertical interconnect technology plays a decisive role in overcoming the limitations of planar integration and significantly improving overall system performance, making it essential for high-density chip stacking and heterogeneous integration [5,6]. Through-silicon vias (TSV) is one of the earliest widely recognized 3D vertical interconnect technologies, noted for its high compatibility with traditional integrated circuit processes and excellent thermal conductivity [7–9]. However, TSV solutions using silicon interposers incur significant fabrication costs. Additionally, the relatively high dielectric constant of silicon adversely affects signal transmission characteristics in high-frequency environments, severely limiting its further application in radio frequency (RF) microsystems with three-dimensional integration [10,11].

Through-glass via (TGV) technology, which involves forming, filling, and metallizing holes in glass substrates, is a key technology for 3D heterogeneous integration. It offers advantages such as excellent dielectric properties, low loss, adjustable coefficient of thermal

This article is part of a special issue entitled: Advanced Packaging published in Materials Science in Semiconductor Processing.

* Corresponding author. State Key Laboratory of Precision Welding & Joining of Materials and Structures, Harbin Institute of Technology, Harbin, 150001, China.

E-mail address: syzhang@hit.edu.cn (S. Zhang).

<https://doi.org/10.1016/j.mssp.2026.110777>

Received 21 February 2026; Received in revised form 27 April 2026; Accepted 7 May 2026

Available online 11 May 2026

1369-8001/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

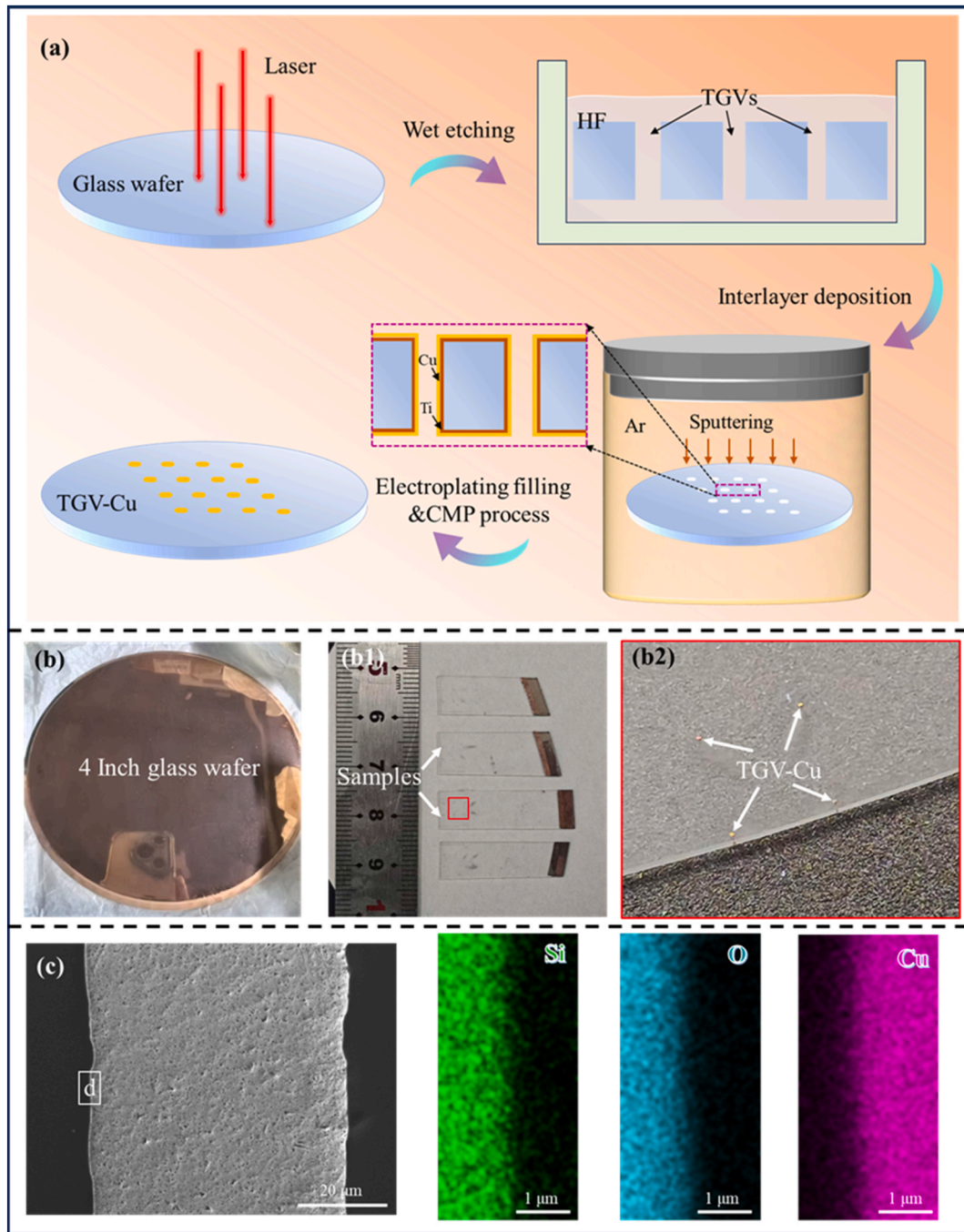


Fig. 1. (a) Schematic illustration of TGV fabrication, metallization, and filling processes on a glass wafer, (b) glass wafer after electroplating, (b1) sample obtained by dicing the plated glass wafer; (b2) enlarged view of a localized region of the diced sample, (c) cross-sectional morphology of the TGV-Cu interconnection structure and the corresponding EDS elemental mapping results in region d.

expansion (CTE), simple processing, and extremely high planarity [12], which impart unique advantages in the manufacturing of high-sensitivity sensors, RF devices, and optoelectronic integrated components [13–15]. Furthermore, the related process can be carried out on large glass substrates (such as 510 mm × 515 mm), facilitating large-scale production, and is considered the next-generation vertical interconnect technology that may replace TSV [16,17]. High-quality metallization filling of glass vias is critical for interchip communication. While some new filling techniques have emerged, electroplating remains the mainstream method [18].

Although both silicon and glass vias are filled by electroplating, their filling mechanisms and interconnect structures differ significantly.

Silicon vias are typically blind holes prepared using the Bosch process, filled from the bottom up after creating insulation and barrier layers, and then followed by Chemical Mechanical Planarization (CMP). In contrast, glass substrates typically have double-sided vias, requiring dual-sided electroplating, resulting in significant differences in residual stress and microstructure [19,20].

Additionally, the two interconnect methods use different substrates—silicon and glass—which differ substantially in physical properties such as CTE, elastic modulus, and thermal conductivity. The interface structures are also distinct: TSV requires an SiO₂ insulating layer, whereas glass itself is insulating and does not require such a layer [21]. Due to differences in processing techniques, silicon vias have

vertical holes with scalloped sidewalls, whereas glass vias often have X-shaped, conical, or slanted holes [22]. Therefore, the hole-formation method, orientation, and interface structure of TGV technology fundamentally differ from those of TSV-Cu interconnects, leading to unique microstructural evolution and interconnect lifetime for TGV-Cu interconnects.

Research has extensively investigated the integrity of electroplated copper within via interconnection structures, specifically TGV-Cu. Okoro et al. [23] investigated the impact of different annealing conditions on the surface cracking patterns of glass substrates, finding that the circumferential stress generated during heating is the primary cause of radial cracks. Wang et al. [24] studied the effect of different annealing times at 400 °C on the protrusion of TGV-Cu and observed the appearance of microcracks after annealing for 1 h. They also developed an elastic-plastic-creep-coupling model to simulate the bump's behavior using finite-element analysis. Zhao et al. [25] and Le et al. [26] analyzed the stress distribution at the Cu/glass interface, providing guidance for process optimization and reliability enhancement. Okoro et al. [27] conducted high-temperature storage, highly accelerated stress testing (HAST), and thermal-shock reliability experiments on glass substrates, focusing on helium hermeticity.

In addition to investigating macro-cracking, copper extrusion, sealing reliability, and stress distribution on glass substrates, Wang et al. [28] employed the $\sin^2\Psi$ method to examine how residual stress evolves within the copper layer of glass interposers as a function of annealing temperature. Their study revealed a transition from compressive to tensile stress when the temperature was increased from 100 °C to 200 °C. The author [29] also systematically examined the effects of heat treatment temperature on the microstructural evolution, electrical resistivity, and elastoplastic constitutive behavior of copper pillars. Presently, research in this domain is predominantly centered on several key aspects: the effects of heat treatment on microstructural and mechanical properties, failure analysis, and the protrusion behavior of TGV-Cu interconnects [30,31]. However, few scholars have thoroughly investigated the microstructure of TGV-Cu interconnects during service, leaving their characterization unresolved. The degradation process and underlying mechanisms remain unclear and urgently require in-depth study.

In this study, aging tests were initially conducted on glass substrates containing TGV-Cu, and the effects of aging time on cross-sectional morphology and elemental diffusion behavior will be subsequently investigated using a scanning electron microscope (SEM) and energy-dispersive spectrometry (EDS). Following this, electron backscatter diffraction (EBSD) will be used to further analyze the microstructure at interface defects within the interconnects. Finally, mechanical property changes are studied using nanoindentation, shedding light on the microstructural degradation and mechanisms of TGV-Cu interconnections.

2. Materials and experiments

2.1. Fabrication process of glass substrate

This study begins by using laser-induced deep etching (LIDE) technology to fabricate glass vias in a 6-inch quartz glass substrate with a thickness of approximately 420 μm [32]. The processing schematic is shown in Fig. 1(a). Initially, a femtosecond laser (Genius-100, Wuhan Huaray Precision Laser Co., Ltd.) is directed to specific locations on a cleaned glass wafer. Laser irradiation induces multiphoton ionization, causing chemical bond breakage and plasma micro-explosions that form a defect-rich, microcracked modified zone within the glass wafer. The modified glass wafer is then immersed in hydrofluoric acid (HF) solution for etching. The chemical reaction equations are presented in Eqs. (1) and (2).



Due to the rapid chemical reactivity and increased specific surface area of the modified region, the etching process exhibits varying reaction rates, allowing holes to be formed in the glass. Subsequently, the prepared glass wafer was placed on the tray of the transfer chamber of the magnetron sputtering system (Polaris B630, Beijing Sevenstar Electronics Co., Ltd.) and then transferred to the main chamber. The vacuum of the main chamber was pumped down to 5×10^{-8} Torr using a mechanical pump and a molecular pump. Once a high-vacuum environment was established, high-purity argon gas (99.999%) was introduced as the working gas. The flow rate was maintained at 30–35 sccm, the working pressure was set to 6 mTorr, the sputter temperature was set to 100 °C, and the tray rotation speed was 25 rpm. After a 20 min pre-sputtering process to clean the titanium and copper targets as well as possible surface contaminants, formal sputtering was carried out. The direct current power was set to 200 W, and under the dual-side sputtering mode, Ti and Cu films with deposition thicknesses of approximately 500 nm and 2 μm were deposited with sputtering times of 1.5 h and 2 h, respectively. The adhesive layer enhances the bond strength between copper and glass, effectively preventing copper diffusion into the glass substrate, while the seed layer provides the necessary adhesion for subsequent copper electroplating.

Afterward, the glass vias are filled via electroplating. The electrolyte used for electroplating consists of copper sulfate, sulfuric acid, and chloride ions, with additives such as sodium sulfate, polyethylene glycol, and brightener. These additives help control the quality and uniformity of copper deposition during electroplating. Finally, CMP is used to remove copper deposited on the glass surface, ensuring the flatness and structural integrity of both the inner and outer vias.

2.2. High-temperature storage experiment

To investigate the failure modes and microstructural evolution of glass substrates under different aging times, this study employed laser cutting to divide four-inch glass wafers into smaller samples, as shown in Fig. 1(b) and (b1). The samples were organized into four groups, each containing five samples, with each sample featuring twelve TGV-Cu interconnect structures. A localized magnification of the TGV-Cu structure is illustrated in Fig. 1(b2). Fig. 1(c) presents the cross-sectional morphology of the initial TGV-Cu interconnect structures, alongside the distribution of Si, O, and Cu elements in region d, as indicated by the white box. The TGV-Cu interconnect interface is continuous and intact, indicating that the fabrication process used in this study is viable.

In accordance with JEDEC standards (JESD22-A103E, Condition F), the high-temperature storage conditions were set to 300 °C, with storage durations of 0, 50, 100, and 200 h. The samples were placed in different crucibles and heated at 10 °C/min to 300 °C, then allowed to cool naturally. The entire experimental procedure was conducted in a muffle furnace (Model: SX-B07123, Tanjin Zhonghuan Electric Furnace Co., Ltd), which maximally simulates the actual service environment of a glass-based device.

2.3. Sample preparation and characterization methods

After aging, the glass samples were encapsulated in molds and subsequently polished with silicon carbide sandpaper grits #800, #2000, and #5000. Once the cross-section of the TGV-Cu was exposed, further polishing was undertaken with a 0.05 μm alumina polishing suspension on a polishing pad to eliminate minor scratches from the TGV-Cu section. To achieve a high-quality Kikuchi pattern, ion polishing was performed on the sample surface using an ion polishing system (Leica EMTIC 3X) to remove the surface stress layer. To enhance the electrical conductivity of the sample surface, a thin gold layer approximately 10 nm thick was sputtered onto the ion-polished samples using an ion

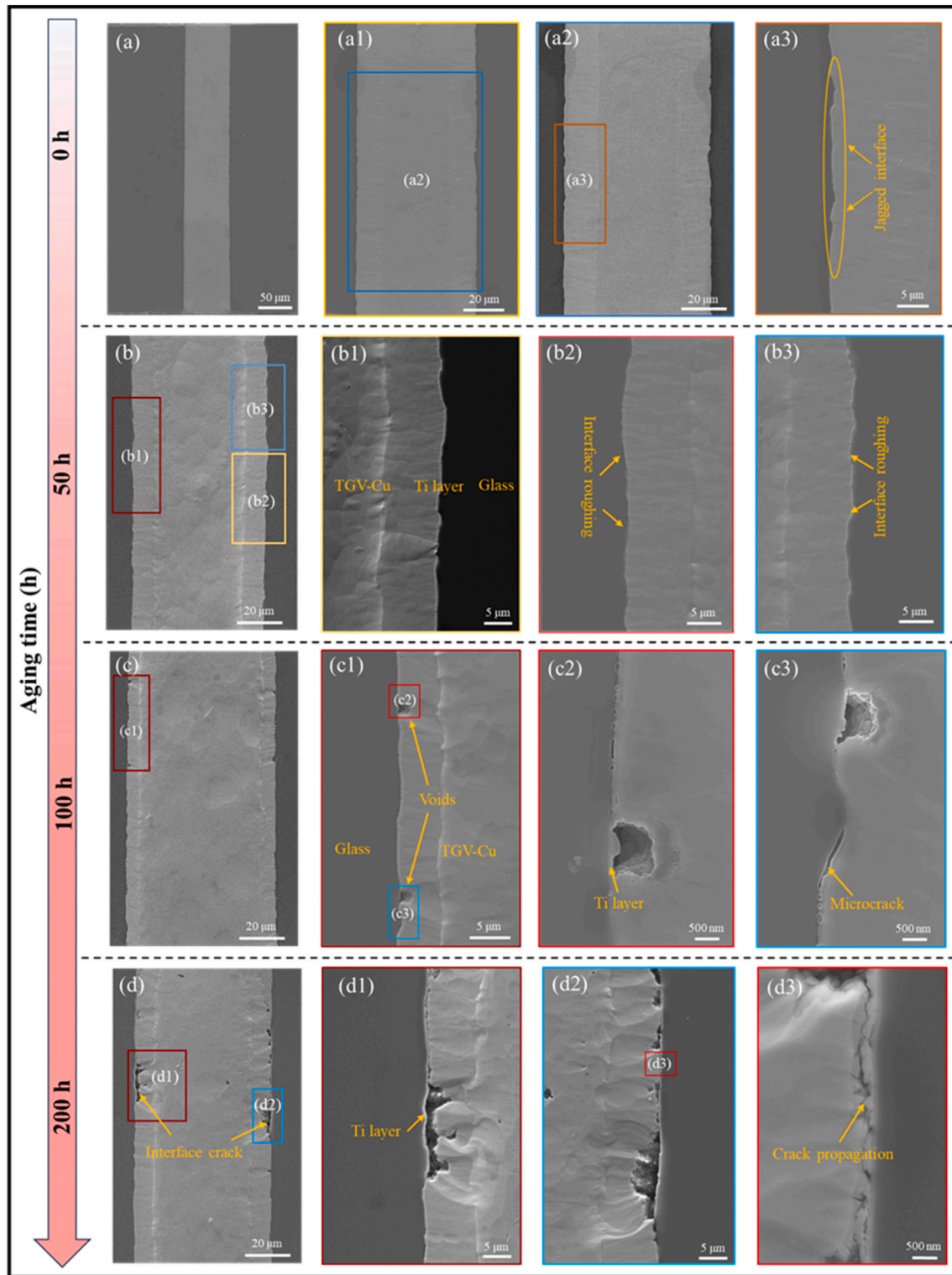


Fig. 2. Morphological evolution of the TGV-Cu interconnection structure cross-section after different aging times: (a-a3) as-electroplated, (b-b3) 50 h, (c-c3) 100 h, (d-d3) 200 h.

sputtering instrument (ISC150T, Supro).

After sample preparation, the microstructure and elemental diffusion behavior of the samples were characterized using a field-emission scanning electron microscope (FESEM, SEM5000, CIQTEK Co., Ltd.) equipped with an electron backscatter diffraction (EBSD) detector (Oxford Symmetry 2) operating at 15 kV. Features such as grain orientation, grain size, and deformation behavior of the samples were captured with

a scanning step size of 150 nm in orientation imaging microscopy mode. Further analysis was conducted using the commercial software AZtec-Crystal 2.1. Ultimately, nanoindentation tests were performed to evaluate the variations in the mechanical properties of TGV-Cu using the Bruker TI980 system. A Berkovich diamond indenter with a radius of 20 nm was utilized, and the indentation depth was set to 500 nm.

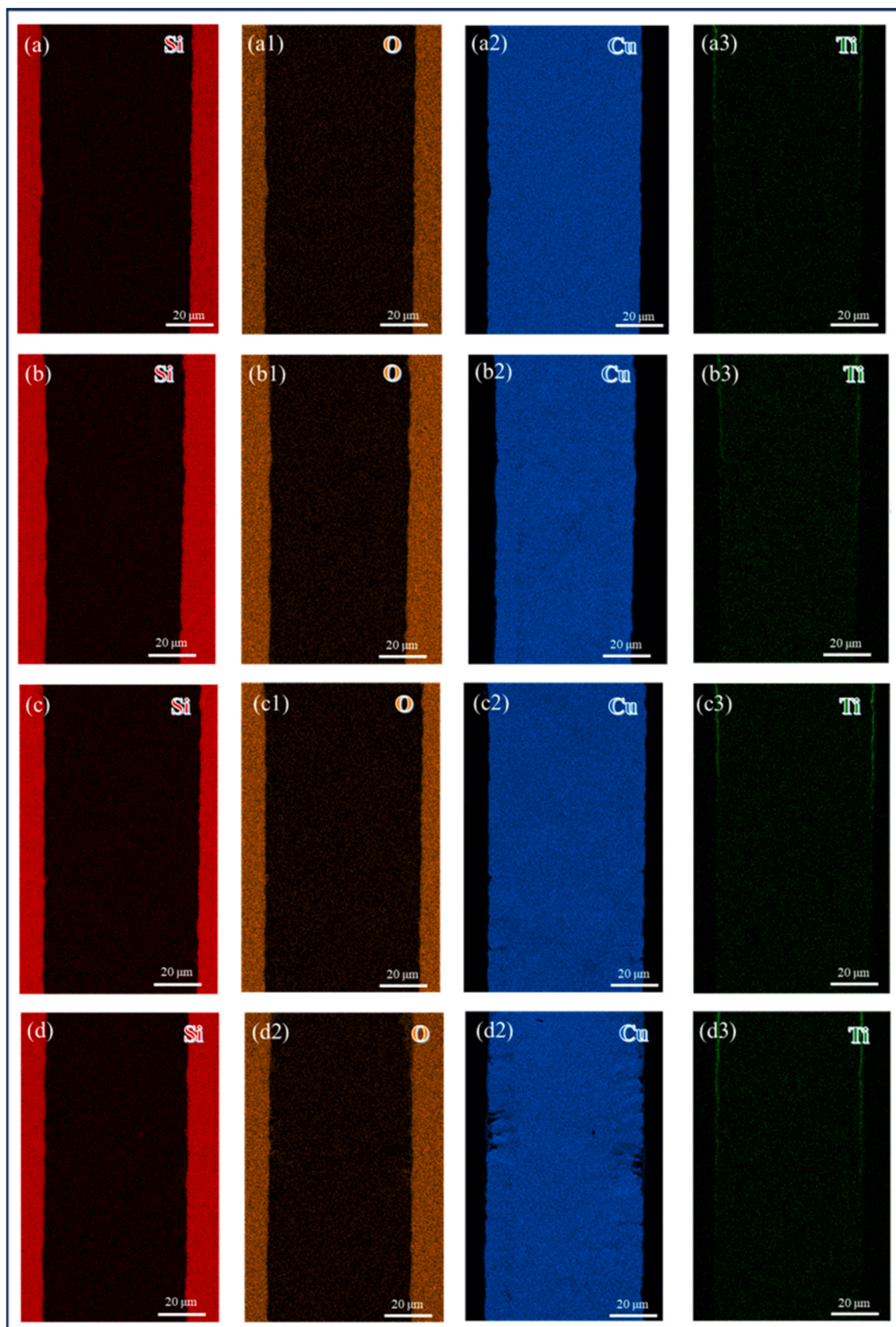


Fig. 3. Elemental distribution of Si, O, Cu, and Ti in the TGV-Cu cross-section after different aging times: (a-a3) as-electroplated, (b-b3) 50 h, (c-c3)100 h, (d-d3) 200 h.

3. Results and discussion

3.1. Effect of aging on interface morphology and element diffusion

Fig. 2 illustrates the cross-sectional morphology of TGV-Cu interconnect structures after varying aging durations. As shown in Fig. 2 (a–a2), the glass vias are fully filled, with a diameter of approximately 62 μm and an aspect ratio of 6:1. The interface between the glass and copper is intact and continuous, devoid of electroplating defects and microcracks. However, due to the differential etching rates of HF acid, the sidewalls of the glass via exhibit a micrometer-scale roughness (1–2 μm), resulting in an uneven interface following electroplating [33], as depicted in Fig. 2(a3).

Fig. 2(b–b3) present SEM-secondary electron images for an aging time of 50 h. The glass/Ti/Cu interface remains continuous without crack initiation, although notable coarsening of the interface occurs. This coarsening is attributed to the interdiffusion of titanium and copper atoms at elevated temperatures, leading to the formation of CuTi_2 and CuTi intermetallic compounds (IMCs), which continue to grow over time [34]. As the aging duration increases, the extent of atomic diffusion increases. After aging for 100 h, microcracks are observed extending along the sidewalls at the Ti/Cu interface. This phenomenon arises from the molar-volume-induced strain generated by the formation of CuTi IMCs, resulting in significant localized tensile stresses that promote crack initiation [35,36].

Additionally, irregularly shaped voids with diameters of approximately 1 μm were observed on the TGV-Cu side, as shown in Fig. 2 (c–c3). According to Ref. [37] and Eq. (3), the molar volume of the preferentially formed CuTi IMC is calculated to be approximately 17.19 cm^3/mol , while the molar volumes of Cu and Ti are 7.18 cm^3/mol and 10.66 cm^3/mol , respectively. Based on Eq. (4), this results in a volume shrinkage of about 3.8% relative to the reactants. This volume contraction generates tensile stress at the interface and creates an excess vacancy flux, thereby providing a strong driving force for the nucleation and growth of Kirkendall voids. Moreover, in the Cu–Ti diffusion couple, Cu is the dominant diffusing species and tends to migrate toward the Ti layer, resulting in the formation of Kirkendall voids on the Cu side [38, 39]. At 200 h of aging, the cross-sectional morphology of TGV-Cu is presented in Fig. 2(d–d3), where the IMC layer has coarsened further. Defects evolve from microcracks and irregular voids into through-thickness cracks extending along the interface, with lengths approaching 20 μm . The core driving force behind this transition in defect mode is attributed to the shear stresses generated at the interface due to the CTE disparity among the glass/Ti/Cu trilayer structure (SiO_2 : 0.5 $\text{ppm}/^\circ\text{C}$, Ti: 11.5 $\text{ppm}/^\circ\text{C}$, and Cu: 16.4 $\text{ppm}/^\circ\text{C}$) [34,40].

$$V_m = \frac{M}{\rho} \quad (3)$$

Where V_m , M , and ρ denote molar volume, molar mass, and density, respectively.

$$\psi = \frac{V_2 - V_1}{V_1} \quad (4)$$

Where ψ denotes volume shrinkage, and V_1 and V_2 denote molar volumes of reactants before and after reaction, respectively.

Consequently, under external and internal stresses, the failure mode of the TGV interconnect structure evolves from microcracks and voids to penetrating cracks along the titanium-copper interface where the IMCs expand. It is noteworthy that during aging, the titanium adhesion layer remains intact and continuous, as shown in Fig. 2(d1), indicating that the bonding strength between titanium and glass exceeds that between titanium and copper.

Fig. 3 depicts the distribution of four elements—Si, O, Cu, and Ti—across the cross-section of the TGV-Cu interface after varying aging durations, corresponding to (a2), (b), (c), and (d) in Fig. 2. Given that

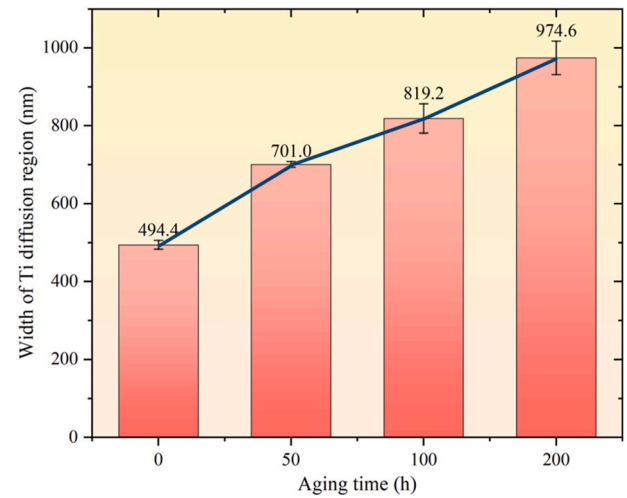


Fig. 4. Plot of the width of the Ti diffusion region as a function of aging time. The error bars represent the maximum-minimum range of the diffusion region width measured at four different locations within the same image.

quartz glass predominantly consists of SiO_2 , the elemental profiles on either side are largely composed of Si and O, while the region of the TGV is primarily occupied by Cu. Additionally, a sputtered Ti seed layer is detected at the glass-Cu interface, indicating a well-defined elemental arrangement in the initial state. Notably, after aging for 50 h, EDS mapping of the TGV cross-section shows no significant deviation from the initial state. However, as aging progresses to 100 and 200 h, the emergence of voids and cracks at the TGV-Cu interface leads to a marked reduction in the Cu distribution, as shown in Fig. 3 (c2) and (d2).

To further investigate the diffusion behavior of the glass-titanium-copper trilayer structure, EDS line scans were performed at two positions on each side of the sample interface, providing a total of four measurement points for each sample. These scans allow for semi-quantitative analysis of the elemental intensities. The variation in the diffusion width of the Ti element over aging time is illustrated in Fig. 4. The initial thickness of the Ti layer is approximately 494.4 nm, and its diffusion width progressively increases with extended aging time. However, the diffusion rate, calculated from the average slope, decreases from an initial value of 4.1 nm/h to 1.5 nm/h over time. The concentration distribution curves of Si, Cu, O, and Ti at the interface are shown in Fig. 5. The diffusion width of the titanium element serves as a quantifiable measure of the intensity of the interfacial reactions. Fig. 5 (a) and (b) indicate that variations in the Ti element are detectable on both sides of the TGV (marked by green lines), revealing that the seed layer widths on either side are generally consistent, which suggests a high level of uniformity in the sputtering process. Furthermore, a rapid decline in the Ti signal is observed in the non-seed layer regions, indicating that Ti atoms have not diffused laterally. After aging for 50 h, the average diffusion width of Ti increases from the initial 494.4 nm to 701.0 nm. At 100 and 200 h, as depicted in Fig. 5(e)–(h), the average diffusion widths of Ti reach 819.2 nm and 974.6 nm, respectively, although the diffusion rate significantly declines.

In a nutshell, the weak points of the TGV-Cu interconnect structure under aging conditions are located at the interface between Cu and Ti. With increasing aging time, the Ti/Cu interface undergoes initial coarsening due to the formation of IMCs, which eventually leads to the development of microcracks and irregularly shaped voids. This progression, driven by thermal stress, results in the gradual propagation of cracks and ultimately leads to through-thickness cracks. Concurrently, there are significant signs of Ti atom diffusion. As the aging time increases, the chemical potential gradient diminishes, resulting in a progressive decrease in the diffusion rate of titanium atoms, which ultimately approaches a state of thermodynamic equilibrium.

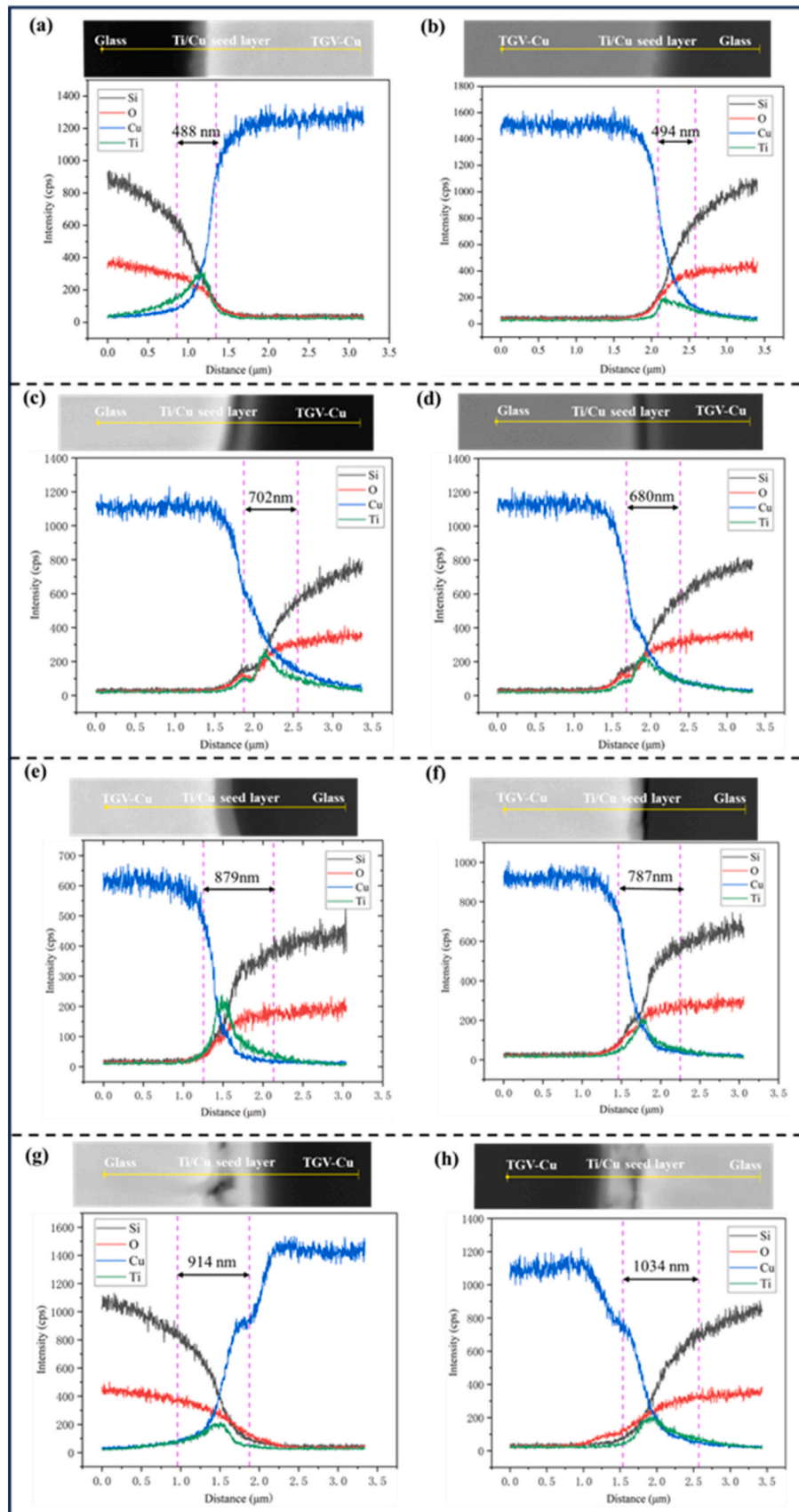


Fig. 5. Elemental concentration distribution curves at the interface of SiO_2 -Ti-Cu multilayer structure after different aging times: (a, b) as-electroplated, (c, d) 50 h, (e, f) 100 h, (g, h) 200 h.

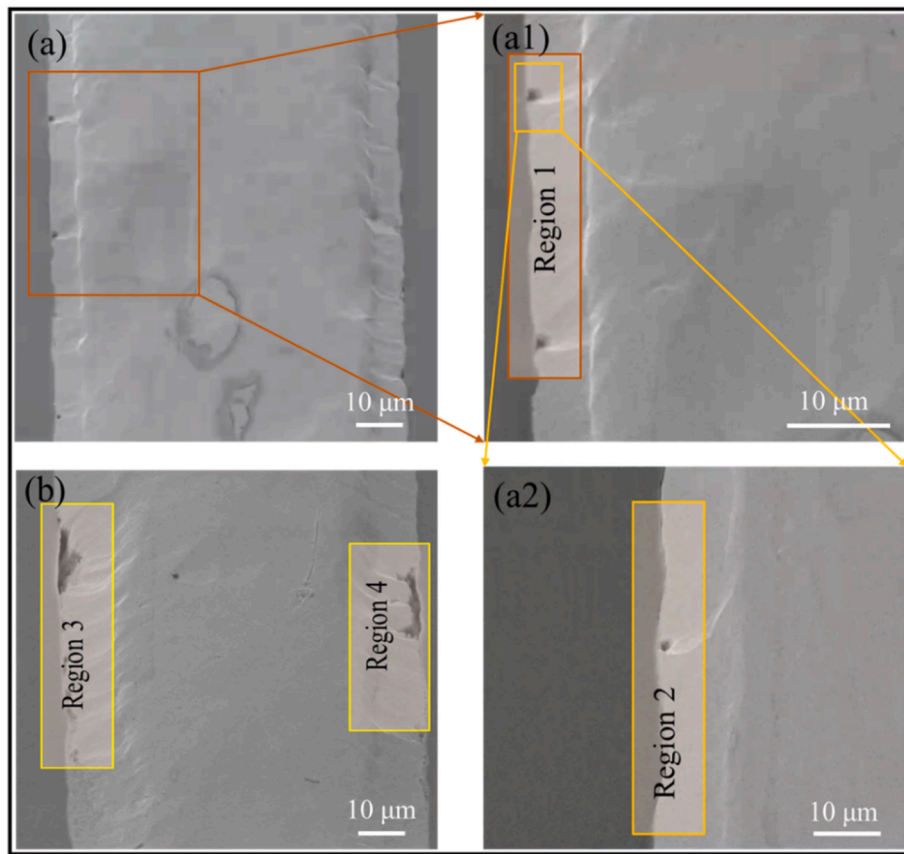


Fig. 6. Evolution of defects from voids to cracks and the corresponding EBSD scanning region.

3.2. Interfacial defect microstructure analysis of TGV-Cu in interconnects

The SEM characterization and EDS analysis results presented in Section 3.1 demonstrate that the evolution of cross-sectional morphology and defects in the TGV-Cu interconnection structure under different aging conditions is highly intriguing. The fracture modes and crack propagation paths of the TGV-Cu interconnection exhibit a close, direct relationship with the evolution of the interface microstructure, which has not been previously observed under aging conditions. This section analyzes the microstructure of the TGV-Cu interconnection at defect sites.

Fig. 6(a-a2) and (b) illustrate the microvoids and cracks observed after aging for 100 h and 200 h, respectively, which have been discussed in Section 3.1. Additionally, the regions labeled as region 1, region 2, region 3, and region 4 in Fig. 6 denote the EBSD sampling areas. The EBSD analysis results for these four regions are shown in Fig. 7. Fig. 7 (a1-a3) and 7 (b1-b3) present the grain orientation distribution maps, kernel average misorientation (KAM) maps, and grain orientation spread (GOS) distribution maps for the void regions, namely regions 1 and 2. It can be observed that the grain size, orientation difference, and grain deformation magnitude are quite consistent between regions 1 and 2. The average GOS values for these two regions are 0.31 and 0.32, both below 2. The GOS value of TGV-Cu less than or equal to 2 is considered to represent recrystallized grains [41]. Therefore, although microvoids induced by diffusion are present at the TGV-Cu interface, the internal strain energy and grain distortion remain relatively low.

Fig. 7 (c1) and (d1) show the grain orientation distribution maps at the crack locations, demonstrating that the cracks primarily propagate along the sidewall direction, resulting in a transgranular fracture mode. This behavior can be attributed to a mismatch in the CTE between glass and copper, along with copper creep, which creates significant shear stress at the interface [29]. The elevated stress regions facilitate fracture

paths that directly traverse the grains. Notably, while the main crack propagates, bifurcating cracks form, extending along the grain boundaries into the interior of the TGV-Cu, as illustrated in Fig. 7(d1). This may be a consequence of the gradual weakening of grain boundaries under prolonged high-temperature conditions, resulting in boundary strengths lower than those of the grains themselves [42]. As shown in Fig. 8, regions 3 and 4—corresponding to the crack locations—exhibit high proportions of high-angle grain boundaries of 90.65% and 81.4%, respectively. The need for additional energy to propagate cracks in these areas leads to greater angular distortion, causing the bifurcating cracks along the grain boundaries to follow a more tortuous path.

The comparative results from Fig. 7(a2, b2) and (c2, d2) indicate that during the evolution process from voids to cracks, the average KAM value at defect sites increases from 0.51 to 0.67, reflecting a 32.37% increase, while the average GOS rises from 0.32 to 0.96, tripling in value. This indicates that residual stress and lattice distortion at defect regions in TGV-Cu increase with aging time, and the distribution becomes increasingly asymmetric. In regions where cracks have already formed (marked by yellow circles), both KAM values and grain deformations are at lower levels due to the release of residual stress, and this resultant space from the cracks permits the grains to deform freely. Conversely, regions without cracks (marked by red circles) demonstrate severe lattice distortion and substantial residual stress, indicating the potential direction for crack propagation.

Therefore, based on the comprehensive comparative analysis presented above, it is concluded that defects transition from voids to cracks, with cracks exhibiting transgranular fracture along the interface and bifurcating to extend along the grain boundaries in a tortuous manner. In regions poised to form new cracks, significant lattice distortion and stress concentration occur. Beyond the interconnect structure interface defects, TGV-Cu itself serves as a primary conduit for electrical signals and heat dissipation. However, the effect of storage time on the

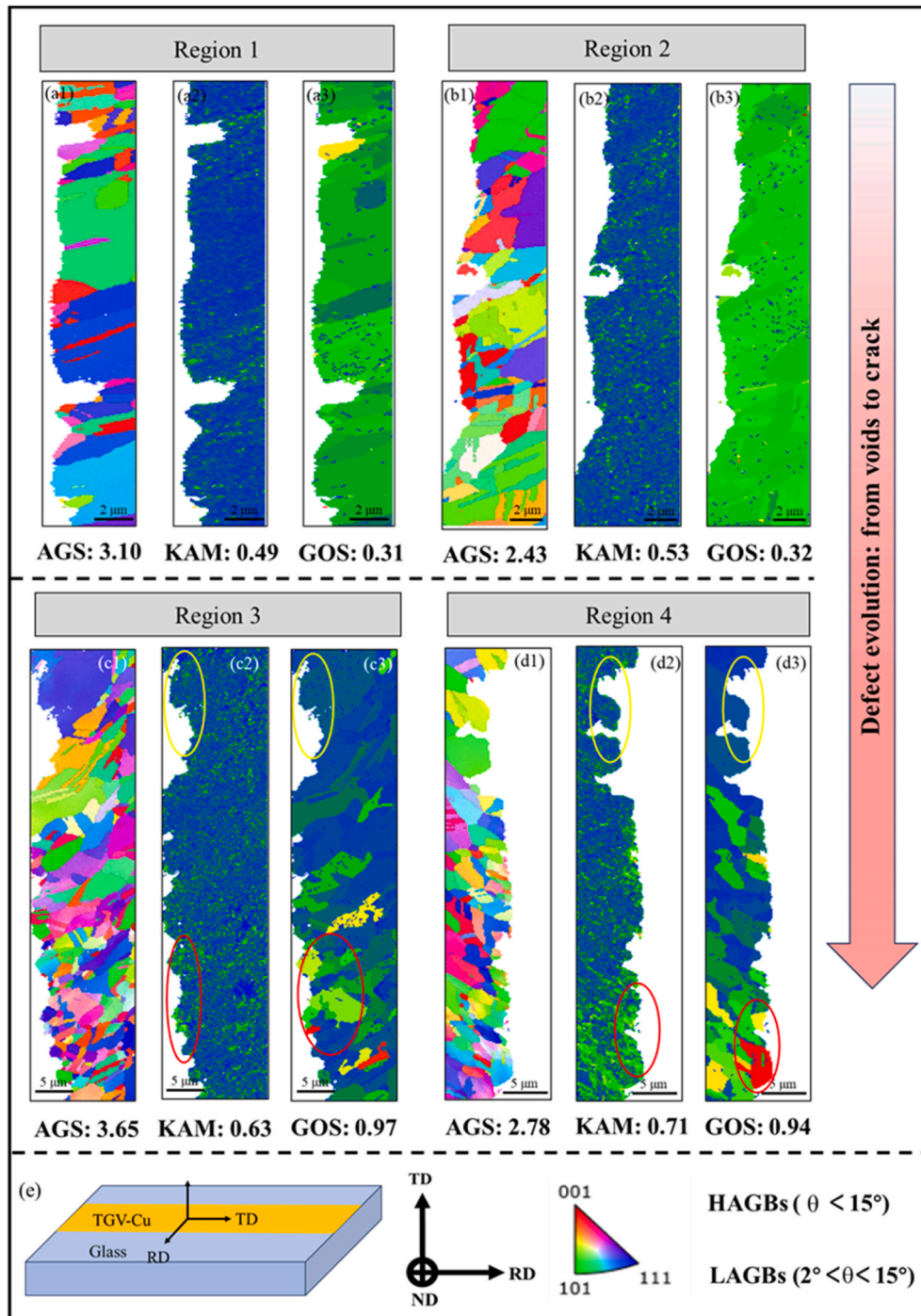


Fig. 7. (a1, b1) and (c1, d1) Grain orientation maps of the void and crack regions, respectively, (a2, b2) and (c2, d2) KAM maps of the void and crack regions, respectively, (a3, b3) and (c3, d3) GOS maps of the void and crack regions, respectively, (e) schematic diagram of the observation zone of the TGV-Cu cross-section. Inset images show the corresponding IPF orientation maps by EBSD.

electroplated copper pillar within glass vias remains unclear; hence, a thorough microstructure analysis will be conducted next.

3.3. Crystallographic changes and evolution of TGV-Cu

3.3.1. Grain size and grain boundary characteristics

To further elucidate the microstructural characteristics of TGV-Cu, EBSD analysis was conducted on TGV-Cu samples aged for various durations. A scanning area of $40 \mu\text{m} \times 80 \mu\text{m}$ was selected in the central region of the TGV-Cu samples. Fig. 9 presents the grain-size distribution

histograms and inverse pole figures obtained by EBSD for TGV-Cu at different aging times. Fig. 9(a) illustrates the grain size distribution of the initial state of TGV-Cu, with an average grain size of $3.13 \mu\text{m}$. The statistical results depicted in Fig. 9(b), (c), and (d) indicate that as aging time increases, the overall grain size distribution shows a pattern with smaller grains at both ends and larger grains in the middle.

After aging for 50 h, the average grain size of TGV-Cu increases to $4.13 \mu\text{m}$, a 31.95% increase from the initial state. When the aging duration is extended to 100 h, the average grain size further increases to $6.41 \mu\text{m}$, a 24.75% rise from the initial state. Finally, after aging for 200

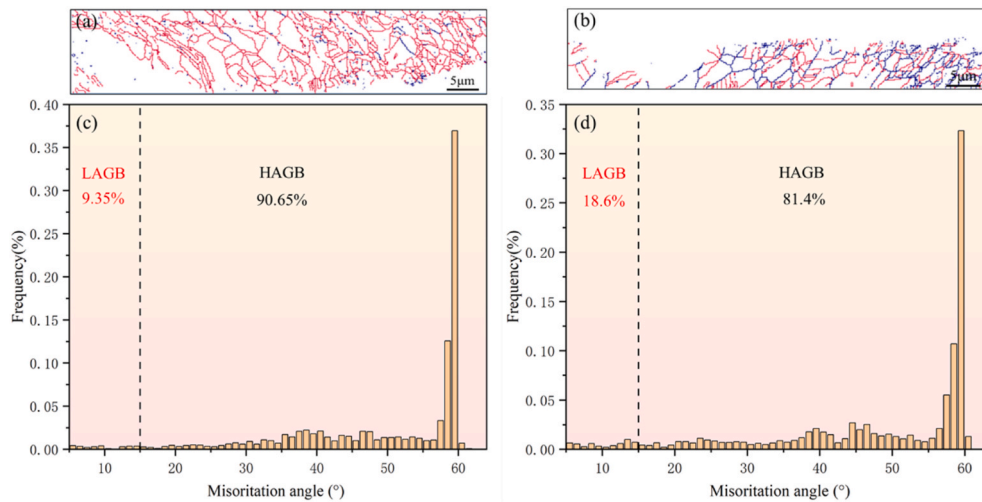


Fig. 8. Grain boundary and the corresponding misorientation angle distribution: (a, c) region 3, (b, d) region 4.

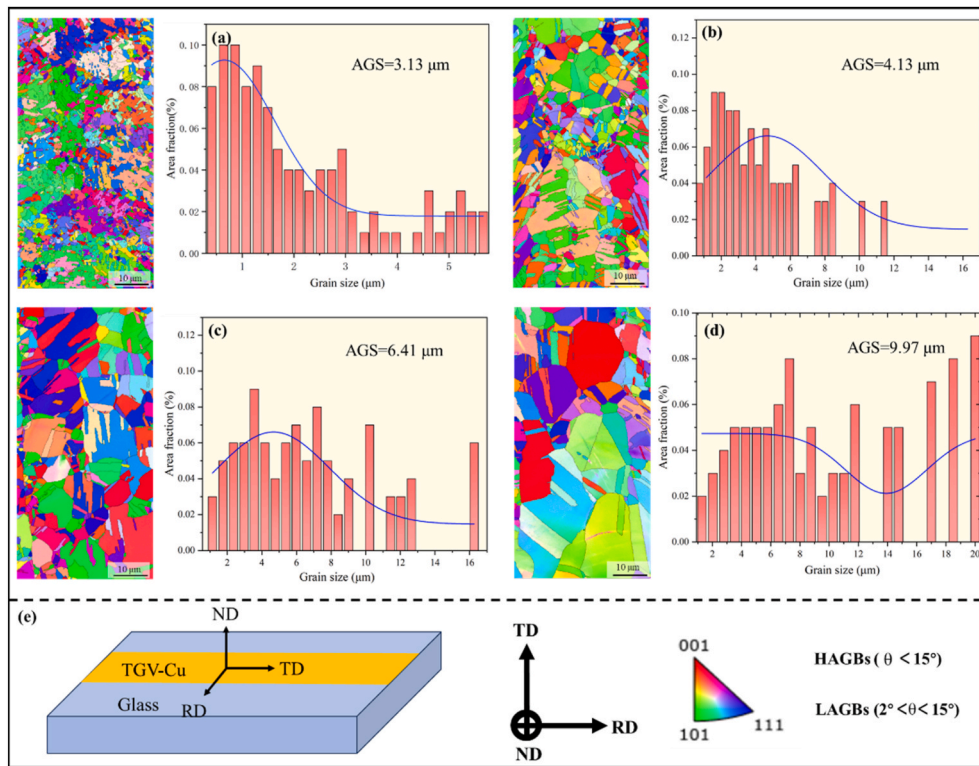


Fig. 9. Grain size distribution histograms in the cross-sections of the TGV-Cu under different storage time: (a) 0 h, (b) 50 h, (c) 100 h, (d) 200 h, (e) schematic diagram of the observation zone of TGV-Cu section. Inset images show the corresponding inverse pole figure orientation maps by EBSD.

h, the average grain size continues to increase, reaching 9.97 μm. Furthermore, the formation of annealing twins within the grains may occur due to the constrained environment of the TGV, where grains seek to accommodate deformation and facilitate adaptation to restrictions by forming twin boundaries, thereby reducing the system's total energy. The occurrence of twinning indicates that, during aging, TGV-Cu likely experiences localized regions of significant strain gradients.

The grain boundary maps obtained from EBSD analysis at different aging times for the TGV-Cu cross-sections are presented in Fig. 10. In the figure, low-angle grain boundaries (LAGBs, $2^\circ \leq \theta \leq 15^\circ$) are represented by gray lines, and high-angle grain boundaries (HAGBs, $\theta > 15^\circ$) are marked with black lines. Initially, the proportion of HAGBs was 93.7%, indicating that most of the recrystallization process had been completed,

consistent with previous characterization of as-deposited TGV-Cu reported in the literature [43]. As aging progressed, an increase in grain size was observed, accompanied by a reduction in the total length of grain boundaries. The statistical results are illustrated in Table 1. After aging for 50 h, the proportion of HAGBs further increased to 98.1%, attributable to residual recovery and recrystallization—a phenomenon that will be validated by KAM mapping. At this stage, TGV-Cu exhibited an almost complete absence of LAGBs. Interestingly, over extended aging, the proportion of HAGBs fluctuated, decreasing to 96.4% and then to 92.2%. This behavior is likely due to abnormal grain growth [44], as evidenced in Fig. 10(c) and (d).

The coincidence site lattice (CSL) consists of sites (atoms) where the lattices of two neighboring grains coincide, and it significantly

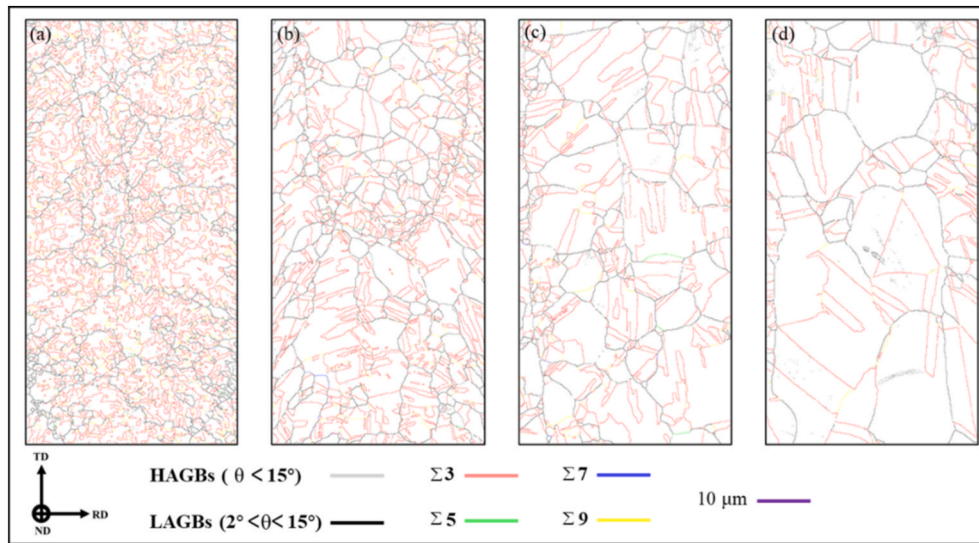


Fig. 10. Grain boundary characteristics of the TGV-Cu: (a) as-electroplated, (b) 50 h, (c) 100 h, (d) 200 h.

Table 1

Proportion of LAGBs and HAGBs after different storage times.

TGV-Cu	0 h	50 h	100 h	200 h
HAGB (%)	93.7	98.1	96.4	92.2
LAGB (%)	6.3	1.9	3.6	8.8

Table 2

Proportion of CSL boundaries after different storage times.

TGV-Cu	0 h	50 h	100 h	200 h
Σ3 (%)	62.6	64.3	62.5	61.5
Σ5 (%)	0.24	0.21	0.71	0.06
Σ7 (%)	0.25	0.43	0.28	0.18
Σ9 (%)	13.1	5.96	5.01	5.08

influences the mechanical properties of polycrystalline materials [45]. For face-centered cubic (FCC) polycrystalline copper characterized by the Fm3m symmetric space group, the primary types of CSL boundaries include Σ3 ($\langle 111 \rangle$ (60°, TB), Σ5 ($\langle 110 \rangle$ 36.9°), Σ7 ($\langle 111 \rangle$ 38.2°), and Σ9, ($\langle 110 \rangle$ 38.9°, TB). The distribution of these CSL boundaries is shown in Fig. 10, with red, green, blue, and yellow lines, and a notable predominance of Σ3 twin boundaries within the grains.

Table 2 illustrates the percentages of Σ3, Σ5, Σ7, and Σ9 boundaries in the TGV-Cu before and after thermal aging, as assessed through EBSD analysis. In each sample, the proportion of Σ3 boundaries is the highest, exceeding 61%. Their prevalence can be attributed to lower energy compared to other boundary types [46]. In this experiment, Σ9 boundaries represent the second most common type, aligning with findings from the work of Wang et al. [47]. As aging time is extended, the proportion of Σ3 boundaries first increases to 64.3%. This behavior can be explained by the relatively low stacking fault energy of copper, which enables twinning to become a principal deformation mechanism in addition to dislocation slip. The Σ3 (60°/ $\langle 111 \rangle$) boundary is recognized as having the lowest energy among boundaries in FCC metals (~ 19 mJ/m²), significantly lower than that of randomly oriented HAGBs at approximately 835 J/m² [48]. Consequently, the release of thermal stress during the thermal aging process preferentially occurs through the formation of Σ3 boundaries via annealing twinning. Under prolonged aging conditions, dislocations rearrange via slip and climb, consuming Σ3 boundaries as nucleation sites for LAGB formation, which is contrary to the trend observed in the LAGB proportion shown in

Table 1.

3.3.2. Grain deformation and crystal orientation

Fig. 11 displays the distribution maps of KAM and GOS for TGV-Cu at different aging times. The statistical results and trends of average KAM and GOS are presented in Table 3. As illustrated in Fig. 11(a1), the initial state of TGV-Cu exhibits a uniformly distributed residual stress, primarily due to the high dislocation density and microstrain introduced during the electrodeposition process. With increasing aging time, the reduction in defect and dislocation densities leads to a decrease in the average KAM value, from an initial 0.6 to 0.39, concurrently with continuous grain growth, as depicted in Fig. 11(a2) and (a3). Interestingly, at an aging time of 200 h, as shown in Fig. 11(a4), the KAM value no longer decreases but instead increases to 0.45.

A similar trend is observed in the variation of the average GOS values, which initially decreases and then increases, as shown in Table 3. This transition is primarily attributed to the recovery and recrystallization processes that occur in TGV-Cu during the early stages of aging. This leads to the formation of new, low-strain grains that grow by consuming surrounding defects, thereby validating the increase in the proportion of HAGBs discussed in Section 3.3.1. However, when the average grain size reaches 9.97 μm, with some grains even attaining sizes of up to 20 μm (approximately one-third of the pore diameter), thermal stresses induced by the constraint of the glass pore walls and the mismatch in CTE lead to an increase in internal strain energy and lattice distortion within the TGV-Cu, particularly in regions of abnormally large grains.

Fig. 12 shows the inverse pole figure (IPF) of the copper pillar in the via before and after aging. The radial direction (RD) and transverse direction (TD) axes represent the radial inverse pole figure, while the normal direction (ND) axis denotes the axial inverse pole figure. In the initial state (Fig. 12(a)), TGV-Cu exhibits a characteristic $\langle 001 \rangle$ preferred orientation [28]. This is attributed to the use of additives and optimized plating parameters during electroplating, aiming for dense, void-free filling, which promotes grain growth along the $\langle 001 \rangle$ direction. Under aging conditions, recrystallization first occurs in regions with high defect density, such as HAGBs and twin boundaries. The orientations of the newly formed grain nuclei tend to be random, influenced by local strain states, and do not necessarily inherit the initial $\langle 001 \rangle$ orientation. Subsequently, these low-strain new grains begin to grow.

For FCC metals such as copper, grains with $\langle 101 \rangle$ and $\langle 111 \rangle$ orientations often exhibit lower surface and strain energies [49,50].

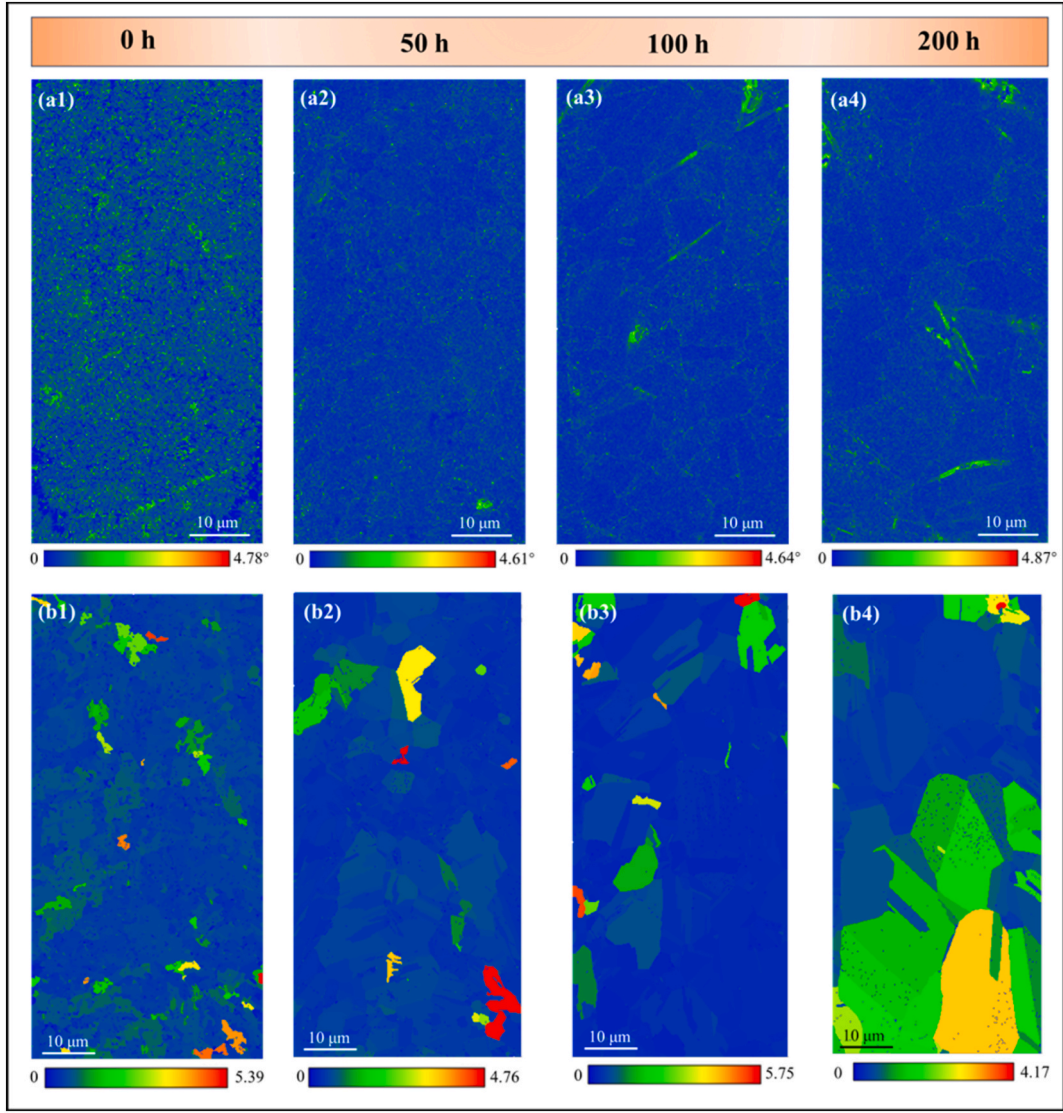


Fig. 11. KAM and GOS distribution maps of the TGV-Cu cross-sections under different storage time: (a1, b1) 0 h, (a2, b2) 50 h, (c1, c2) 100 h, (d1, d2) 200 h.

Table 3

Average KAM and GOS values of TGV-Cu after different storage times.

TGV-Cu	0 h	50 h	100 h	200 h
Average KAM (°)	0.6	0.43	0.39	0.45
Average GOS (°)	0.68	0.57	0.52	0.45

Consequently, as aging progresses, the texture transitions from $\langle 001 \rangle$ to $\langle 101 \rangle$ and $\langle 111 \rangle$, as shown in Fig. 12(b). However, as aging time increases, the geometric constraint effects become pronounced, leading to higher shear stresses that induce grain rotation. Therefore, under the combined influence of strong geometric constraints and the memory of the original texture, $\langle 001 \rangle$ re-establishes itself as the most stable preferred orientation, leading to a return of the texture to its initial state, as depicted in Fig. 12(c) and (d).

3.4. Mechanical properties changes of TGV-Cu interconnects

The mechanical properties of TGV-Cu were assessed by nanoindentation after aging, performed on the region of TGV-Cu corresponding to the area scanned by EBSD in Section 3.3. Five measurement points were taken for each sample to ensure reliability and minimize

errors. Fig. 13(a) shows a typical load-displacement curve of the nano-indenter during loading and unloading. As the indenter is loaded, the load increases progressively with displacement, with the maximum load $f_{\max}$ corresponding to the maximum displacement $h_{\max}$ during the loading phase. The indenter induces both elastic and plastic deformation in the TSV-Cu during loading. Upon unloading, the elastic deformation recovers, while plastic deformation results in a residual displacement h_r in the surface indentation. The contact depth h_c is the indenter's penetration into the material. Nanoindentation testing was performed on TGV-Cu after different aging durations, yielding load-displacement curves as shown in Fig. 13(c–f). Using the Oliver-Pharr method [51], the hardness and elastic modulus of TGV-Cu were calculated from Eqs. (5) and (5).

$$H = \frac{P_{\max}}{S} \quad (5)$$

Where H is the hardness (GPa); $P_{\max}$ is the maximum load (μN); A is the projection of the indentation area (nm^2).

$$\frac{1}{E_r} = \frac{1 - \nu^2}{E} + \frac{1 - \nu_i^2}{E_i} \quad (6)$$

Where E_r and E_i represent the equivalent elastic modulus of TGV-Cu and

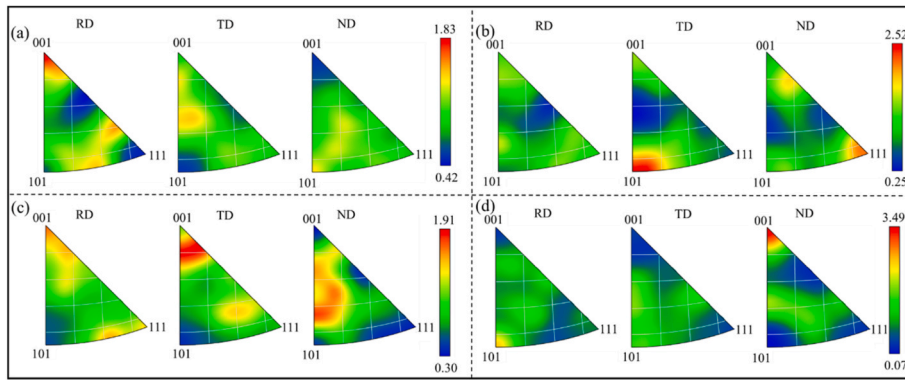


Fig. 12. IPF map of the TGV-Cu (a) as-electroplated, (b) 50 h, (c) 100 h, (d) 200 h.

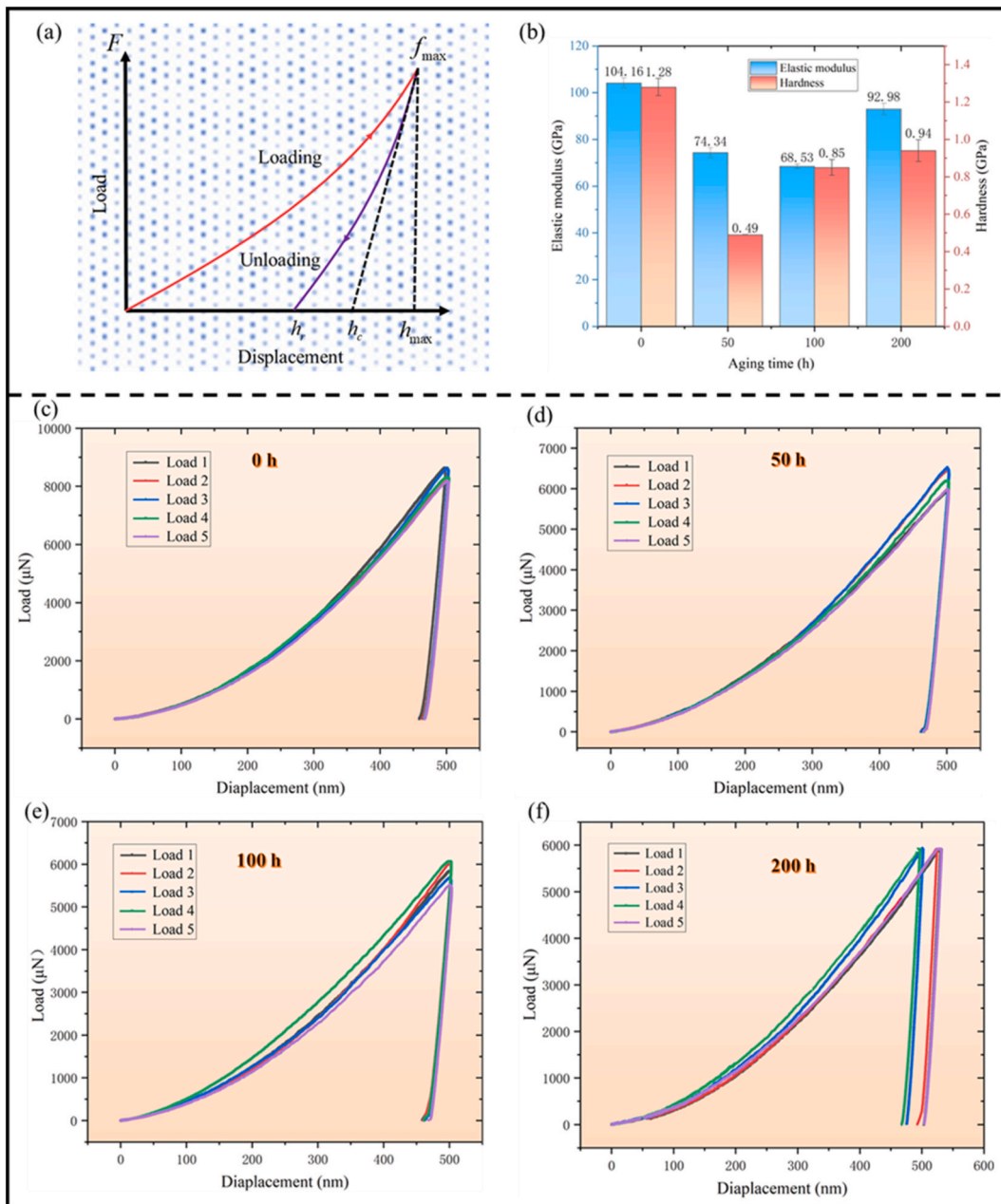


Fig. 13. (a) Schematic diagram of the nanoindentation testing process, (b) elastic modulus and hardness of TGV-Cu after different thermal aging durations, (c-f) displacement-load curves of TGV-Cu corresponding to different aging times.

the elastic modulus of the Berkovich indenter, respectively; ν and ν_i represent the Poisson's ratio of TGV-Cu and the Poisson's ratio of the Berkovich indenter, respectively.

The nanoindentation test results (Fig. 13(b)) reveal a non-monotonic evolution in the elastic modulus and hardness of TGV-Cu during thermal aging. As aging time increases, both properties initially decrease, then gradually recover. However, the overall performance remains lower than the initial state. In the unaged condition, the average elastic modulus and hardness of the material are 104.16 GPa and 1.28 GPa, respectively. After aging for 50 h, both properties decrease to 74.34 GPa and 0.49 GPa, respectively. Based on the EBSD analysis, the softening observed during this phase is primarily attributed to two factors: firstly, the aging process promotes grain growth in TGV-Cu, reducing the total number of grain boundaries and thereby lowering the resistance to dislocation motion. Secondly, the recovery and recrystallization processes occurring early in the aging period effectively release the internal residual stresses.

With continued aging beyond 50 h, both the elastic modulus and hardness gradually recover to 92.98 GPa and 0.94 GPa, respectively. This recovery is primarily associated with texture evolution. EBSD analysis (Fig. 12) confirms that after aging for 50 h, the preferred orientation of TGV-Cu progressively shifts toward the $\langle 001 \rangle$ direction. Although grain coarsening continues and the total grain boundary length decreases, the dominant influence of the strong $\langle 001 \rangle$ texture on mechanical behavior results in an increase in both elastic modulus and hardness. However, these values remain lower than the initial ones. This is because, although the preferred orientation of TGV-Cu is restored to $\langle 001 \rangle$, the average grain size at this stage reaches 9.97 μm , which is larger than the initial average grain size.

3.5. To enhance the reliability of TGV-Cu interconnects

As indicated by the aforementioned research, it appears challenging to mitigate titanium diffusion and prevent the formation of voids and cracks during the aging process. According to Section 3.1, defects initially appear at interfaces, suggesting that reliability could be improved by addressing the interconnect interface structure. For instance, modifying the types and deposition methods of adhesion and seed layers could be beneficial [52]. Furthermore, the introduction of a stress buffer layer or the use of annular vias may reduce the risk of cracking [53]. These strategies could partially mitigate defect formation and may provide guidance for our future investigations.

4. Conclusions

In this study, the glass/Ti/Cu interface gradually coarsened after aging for 50 h. After 100 h, irregularly shaped voids and microcracks appeared at the Cu/Ti interface and progressively propagated into through-thickness cracks driven by thermal stress. Crack propagation occurred predominantly along the sidewall direction and was characterized by a transgranular fracture mode. Moreover, as the primary crack advanced, secondary branching cracks were initiated, intermittently extending into the interior of the TGV-Cu structure along grain boundaries. With aging progressed, the grain size increased from 3.12 μm to 9.97 μm , displaying spatial heterogeneity. Over 92% of the grain boundaries were high-angle, with $\Sigma 3$ boundaries forming the majority of the coincidence site lattice. These $\Sigma 3$ boundaries acted as nucleation sites for low-angle grain boundaries under thermal stress. Initially, the preferred orientation was $\langle 001 \rangle$, which shifted to $\langle 101 \rangle$ and $\langle 111 \rangle$ after aging for 50 h. However, as aging progressed, the $\langle 001 \rangle$ orientation re-emerged as dominant due to the combined effects of geometric constraints and texture memory. In the initial state, the average elastic modulus and hardness of the material are 104.16 GPa and 1.28 GPa, respectively.

CRediT authorship contribution statement

Guoli Sun: Writing – original draft, Investigation, Formal analysis, Data curation. **Jinhong Liu:** Writing – original draft, Data curation. **Jingyi Zhao:** Supervision, Resources. **Jialin Zhang:** Software, Investigation. **Zhaoyu Li:** Investigation, Data curation. **Shuye Zhang:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the open research fund of Songshan Lake Materials Laboratory (No. 2023SLABFN33).

Data availability

Data will be made available on request.

References

- [1] S. Zhang, Z. Li, H. Zhou, R. Li, S. Wang, K.-W. Paik, P. He, Challenges and recent perspectives of 3D heterogeneous integration, *E-Prime - advances in Electrical Engineering, Electronics and Energy* 2 (2022) 100052.
- [2] X. Jing, W. Sun, Y.-H. Wong, P. He, L. Cao, S. Zhang, Stress-Driven reliability-by-design for heterogeneous integration: from packaging architectures to intelligent management, *Chip* (2026) 100195.
- [3] X. Chen, S. He, K.-W. Paik, Y.-H. Wong, S. Zhang, Machine learning-driven design and optimization of electronic packaging: applications and future developments, *J. Mater. Inf.* 5 (2025).
- [4] Y. Xia, Y. Su, X. Xu, L. Ju, R. Zhang, Y. Xue, Y. Liu, S. Zhang, Effect of thermal cycling on the strain and stress distribution of TSVs and solder bumps in stacked package structure, *E-Prime - advances in Electrical Engineering, Electronics and Energy* 5 (2023) 100274.
- [5] W.-Y. Woon, A. Kasperovich, J.-R. Wen, K.K. Hu, M. Malakoutian, J.-H. Jhang, S. Vaziri, I. Datye, C.C. Shih, J.F. Hsu, X.Y. Bao, Y. Wu, M. Nomura, S. Chowdhury, S.S. Liao, Thermal management materials for 3D-stacked integrated circuits, *Nat. Rev. Electr. Eng.* 2 (2025) 598–613.
- [6] H. Zhao, M. Chen, Y. Peng, Q. Wang, M. Kang, L. Cao, TXV Technology: the cornerstone of 3D system-in-packaging, *Sci. China Technol. Sci.* 65 (2022) 2031–2050.
- [7] M. Tian, X. Gu, Alternative insulation liners for through-silicon vias: a comprehensive review, *Mater. Sci. Semicond. Process.* 166 (2023) 107726.
- [8] W. He, Z. Wang, J. Li, Q. Li, Investigation of heat transfer performance for through-silicon via embedded in micro pin fins in 3D integrated chips, *Int. J. Heat Mass Tran.* 214 (2023) 124442.
- [9] B.C. Wolz, C. Jaremenko, F. Vollnhals, L. Kling, J. Wrege, S. Christiansen, X-ray microscopy and automatic detection of defects in through silicon vias in three-dimensional integrated circuits, *Eng. Rep.* 4 (2022) e12520.
- [10] T. Chaloun, S. Brandl, N. Ambrosius, K. Kröhnert, H. Maune, C. Waldschmidt, RF glass technology is going mainstream: review and future applications, *IEEE J. Microw.* 3 (2023) 783–799.
- [11] U. Shah, J. Liljeholm, J. Campion, T. Ebeffors, J. Oberhammer, Low-loss, high-linearity RF interposers enabled by through glass vias, *IEEE Microw. Wireless Compon. Lett.* 28 (2018) 960–962.
- [12] Z. Fang, J. Zhang, J. Liu, J. Fu, S. Li, X. Cai, L. Gao, H. Chen, W. Zhang, Exploration and analysis of through-glass vias for high-speed, low-loss vertical interconnects in glass-based 3-D integrated circuits, *IEEE Trans. Electron. Dev.* 71 (2024) 4267–4274.
- [13] C. Yu, S. Wu, Y. Zhong, R. Xu, T. Yu, J. Zhao, D. Yu, Application of through glass via (TGV) technology for sensors manufacturing and packaging, *Sensors* 24 (2023) 171.
- [14] M.A. Khan, A. Ullah, Z.-J. Fu, S. Khan, S. Khan, Image restoration via combining a fractional order variational filter and a TGV penalty, *Multimed. Tool. Appl.* 83 (2024) 60393–60418.
- [15] C. Ge, X. Wang, J. Du, X. Lin, T. Lei, L. Du, Z. He, High-speed wafer-level TGV interposer for 2.5D CPO, *Opt. Commun.* 579 (2025) 131517.
- [16] V. Sukumaran, T. Bandyopadhyay, V. Sundaram, R. Tummala, Low-cost thin glass interposers as a superior alternative to silicon and organic interposers for packaging of 3-D ICs, *IEEE Trans. Compon. Packag. Manuf. Technol.* 2 (2012) 1426–1433.
- [17] M. Letz, T. Gotschke, F. Wagner, M. Heiss-Chouquet, L. Müller, U. Peuchert, D. Vanderpool, Structured glass substrates in wafer- and panel level packaging:

- status and recent achievements, in: International Symposium on Microelectronics 2021, 2021, pp. 98–102.
- [18] Z.-J. Ji, H.-Q. Ling, P.-L. Wu, R.-Y. Yu, D.-Q. Yu, M. Li, Development status of copper electroplating filling technology in through glass via for 3D interconnections, *J. Electrochem.* 28 (2022) 9.
 - [19] C. Liang, Y. Chen, Y. Guo, W. Wu, M. Hou, H. Liu, L. Ma, X. Chen, C. Wong, Asymmetric bidirectional pulse electrodeposition enabling super-filling of through glass vias with large aperture and high aspect ratio, *IEEE Trans. Compon. Packag. Manuf. Technol.* 15 (2025) 1117–1125.
 - [20] B. Jiao, J. Qiao, S. Jia, R. Liu, X. Wei, S. Yun, Y. Kong, Y. Ye, X. Du, L. Yu, B. Cong, Low stress TSV arrays for high-density interconnection, *Eng. Times* 38 (2024) 201–208.
 - [21] V. Jousseau, C. Guerin, K. Ichiki, M. Lagrange, B. Altemus, C. Zavvou, M. Veillerot, T. Mourier, J. Faguet, Wafer scale insulation of high aspect ratio through-silicon vias by iCVD, *ACS Appl. Mater. Interfaces* 16 (2024) 31624–31635.
 - [22] Z. Yu, Y. Gao, X. Lei, Y. Chen, K. Xu, Y. Zhou, F. Zhu, Effect of tilted wall in through glass via on the electrochemical deposition process, *Electrochim. Acta* 533 (2025) 146556.
 - [23] C. Okoro, S. Jayaraman, S. Pollard, Understanding and eliminating thermo-mechanically induced radial cracks in fully metallized through-glass via (TGV) substrates, *Microelectron. Reliab.* 120 (2021) 114092.
 - [24] H. Wang, J. Xiang, B. Ma, F. Tian, X. Zhong, X. Jian, S. Chen, H. Chen, X. Yang, Modeling and characterization of annealing effect on the Cu protrusion and thermomechanical behaviors of through glass via, *IEEE Trans. Compon. Packag. Manuf. Technol.* 14 (2024) 1300–1307.
 - [25] J. Zhao, Z. Chen, F. Qin, D. Yu, Thermo-mechanical reliability study of through glass vias in 3D interconnection, *Micromachines* 13 (2022) 1799.
 - [26] X.-B. Le, S.-H. Choa, A comprehensive numerical analysis for preventing cracks in 2.5D glass interposer, *J. Mech. Sci. Technol.* 38 (2024) 3027–3037.
 - [27] C. Okoro, P. Maurey, R. Davis, S. Pollard, Evaluation of the helium hermeticity reliability of copper through-glass vias, *Microelectron. Reliab.* 137 (2022) 114783.
 - [28] M. Wang, J. Zhang, L. Gao, H. Chen, W. Li, J. Fu, L. Zhao, D. Wang, S. Li, T. Liu, X. Cai, Y. Li, Bin Peng, W. Zhang, Effect of heat treatment processes on the Cu-electrodeposited through glass vias (TGV) plate, *J. Alloys Compd.* 997 (2024) 174928.
 - [29] M. Wang, J. Zhang, L. Gao, H. Chen, W. Luo, W. Li, X. Han, D. Wang, S. Li, T. Liu, X. Cai, Y. Li, B. Peng, W. Zhang, The mechanical properties and microstructural evolution of copper pillar in through glass vias under different temperatures, *ACS Appl. Electron. Mater.* 7 (2025) 8954–8968.
 - [30] Y. Lai, K. Pan, S. Park, Thermo-mechanical reliability of glass substrate and through glass vias (TGV): a comprehensive review, *Microelectron. Reliab.* 161 (2024) 115477.
 - [31] K. Pan, J. Yang, Y. Lai, S. Park, C. Okoro, D. Joshi, S. Pollard, High-temperature constitutive behavior of electroplated copper TGV through numerical simulation, *IEEE Trans. Compon. Packag. Manuf. Technol.* 13 (2023) 1861–1867.
 - [32] L. Chen, D. Yu, Investigation of low-cost through glass vias formation on borosilicate glass by picosecond laser-induced selective etching, *J. Mater. Sci. Mater. Electron.* 32 (2021) 16481–16493.
 - [33] Z. Fang, J. Zhang, J. Liu, H. Chen, L. Gao, X. Yang, W. Li, X. Cai, H. Guo, Analysis and optimization of sidewall roughness on microwave performance of through-glass vias in 3-d integrated circuits, *IEEE Trans. Microw. Theor. Tech.* 72 (2024) 54–63.
 - [34] Y. Wei, L. Zhu, Y. Li, Y. Chen, B. Guo, Formation mechanism and microstructure evolution of Cu/Ti diffusion bonding interface and its influence on joint properties, *Vacuum* 213 (2023) 112167.
 - [35] Y.K. Kim, H. Pouraliakbar, S.I. Hong, Effect of interfacial intermetallic compounds evolution on the mechanical response and fracture of layered Ti/Cu/Ti clad materials, *Mater. Sci. Eng.* 772 (2020) 138802.
 - [36] C. Wang, W. Chen, Z. Lu, Y. Huang, F. Zhang, Y. Zhang, L. Wang, C. Zhou, S. Cao, K. Yang, Z. Zhou, J. Zhao, D. Sun, F. Meng, D. Ma, Cascaded intermetallic compound formation in additively manufactured Ti-Cu high-concentration alloys, *Intermetallics* 188 (2026) 109087.
 - [37] S. Chen, Y.-H. Duan, B. Huang, W.-C. Hu, Structural properties, phase stability, elastic properties and electronic structures of Cu–Ti intermetallics, *Philos. Mag.* 95 (2015) 3535–3553.
 - [38] S. Ma, L. Chen, T. Zhan, Y. Wang, Y. Wang, Q. Wang, X. Xiao, W. Wang, TGV through-hole double-sided electroplated copper filling mode regulation and process reliability test study, *Scientia Sinica Chimica* 53 (2023) 2068–2078.
 - [39] H. Guo, M. Rao, J. Zhang, X. Wang, G. Luo, Q. Shen, Electromigration-enhanced Kirkendall effect of Cu/Ti direct diffusion welding by sparking plasma sintering, *J. Mater. Process. Technol.* 315 (2023) 117933.
 - [40] Y. Zhao, X. Lu, J. Wang, B. Qi, Z. Hu, Y. Hu, G. Zeng, Z. Lin, R. Sun, SiO₂/SiC binary systems reinforced epoxy resins with high thermal conductivity and low CTE as underfill for 2.5D/3D electronic packaging, *Compos. Commun.* 56 (2025) 102353.
 - [41] L. Shi, J. Zou, Z. Zhang, T. Dang, S. Liang, L. Sun, L. Zhu, J. Cheng, W. Han, D. Song, Y. Wang, Y. Jiang, R. Fei, Y. Song, Controllable phase transformation and static recrystallization inducing excellent strength-plasticity synergy in powder sintered Cu-18Sn-0.3Ti alloy, *J. Mater. Sci. Technol.* 249 (2026) 67–81.
 - [42] A. Wimmer, M. Smolka, W. Heinz, T. Detzel, W. Robl, C. Motz, V. Eyert, E. Wimmer, F. Jahnle, R. Treichler, G. Dehm, Temperature dependent transition of intragranular plastic to intergranular brittle failure in electrodeposited Cu micro-tensile samples, *Mater. Sci. Eng.* 618 (2014) 398–405.
 - [43] Y.-H. Chang, Y.-M. Lin, C.-Y. Lee, P.-C. Hsu, C.-M. Chen, C.-E. Ho, Through glass via (TGV) copper metallization and its microstructure modification, *J. Mater. Res. Technol.* 31 (2024) 1008–1016.
 - [44] P.J. Noell, E.M. Taleff, Subgrains, texture evolution, and dynamic abnormal grain growth in a Mo rod material, *Metall. Mater. Trans. A* 50 (2019) 4608–4619.
 - [45] Z. Gao, R. Hu, Z. Huang, Y. Wu, J. Li, M. Zhou, Metastable transformation behavior in a Ta-containing TiAl-Nb alloy during continuous cooling, *J. Alloys Compd.* 904 (2022) 164088.
 - [46] S. Kobayashi, R. Kobayashi, T. Watanabe, Control of grain boundary connectivity based on fractal analysis for improvement of intergranular corrosion resistance in SUS316L austenitic stainless steel, *Acta Mater.* 102 (2016) 397–405.
 - [47] M. Wang, J. Zhang, L. Gao, H. Chen, W. Li, J. Fu, L. Zhao, Y. Qiao, K. Qin, D. Wang, S. Li, T. Liu, X. Cai, Y. Li, B. Peng, W. Zhang, The influence of heat treatment process on electroplated metallized through glass vias (TGV) substrates, *J. Alloys Compd.* 1035 (2025) 181522.
 - [48] A.J. Birnbaum, J.C. Steuben, E.J. Barrick, A.P. Iliopoulos, J.G. Michopoulos, Intrinsic strain aging, Σ3 boundaries, and origins of cellular substructure in additively manufactured 316L, *Addit. Manuf.* 29 (2019) 100784.
 - [49] M. Stangl, M. Lipták, A. Fletcher, J. Acker, J. Thomas, H. Wendrock, S. Oswald, K. Wetzig, Influence of initial microstructure and impurities on Cu room-temperature recrystallization (self-annealing), *Microelectron. Eng.* 85 (2008) 534–541.
 - [50] J.-M. Zhang, K.-W. Xu, V. Ji, Dependence of strain energy on the grain orientations in an FCC-polycrystalline film on rigid substrate, *Appl. Surf. Sci.* 185 (2002) 177–182.
 - [51] W.C. Oliver, G.M. Pharr, An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments, *J. Mater. Res.* 7 (1992) 1564–1583.
 - [52] F. Yang, Q. Wang, J. Lee, Y. Song, I. Hwang, S. Yoon, B. Yoo, Silicon dioxide (SiO₂) solid blocking layer on through-glass via surface for efficient highly selective copper filling, *Surf. Interfaces* 55 (2024) 105431.
 - [53] O. Ahmed, C. Okoro, S. Pollard, T. Jiang, The effect of materials and design on the reliability of through-glass vias for 2.5D integrated circuits: a numerical study, *Multidiscip. Model. Mater. Struct.* 17 (2020) 451–464.